Space Missions

I] Textbook Questions:

⭐ 1 Mark Objectives: ⭐

1. State with reasons whether the following sentences are true or false

a. If a spacecraft has to be sent away from the influence of earth's gravitational field, its velocity must be less than the escape velocity

Ans: False. A spacecraft must attain escape velocity or greater to escape the Earth's gravitational field.

b. **The escape velocity on the moon is less than that on the earth.** [2025]

Ans: True. The escape velocity on the Moon is less than that on the Earth due to its weaker gravity.

c. A satellite needs a specific velocity to revolve in a specific orbit.

Ans: True: A satellite needs a specific velocity called critical velocity to revolve in a specific orbit.

d. If the height of the orbit of a satellite increases, its velocity must also increase.

Ans: False: As the height of a satellite's orbit increases, its critical velocity decreases, so the orbital velocity decreases.

2. Fill in the blanks and explain the statements with reasoning:

a. If the height of the orbit of a satellite from the earth surface is increased, the tangential velocity of the satellite will **decrease**.

Ans: Because the distance between the satellite and the Earth's increases. Hence, the tangential velocity called the critical velocity of the satellite will decrease. It is given by formula $V_c = \sqrt{\frac{GM}{R+h}}$

b. The initial velocity (during launching) of the Mangalyaan, must be greater than **escape velocity** of the earth. [2019]

Ans: Because the minimum velocity needed by an object to escape earth gravitational field is the escape velocity. It is given by the formula, $V_{esc} = \sqrt{\frac{2GM}{R}}$ [2022]

⭐ 2 Marks Questions: ⭐

1. Why are geostationary satellites not useful for studies of polar regions? [2019]

Ans:

1. Geostationary satellites are satellites that orbit above the Earth's equator.
2. Their time of revolution and the direction of rotation are the same as that of the Earth. These satellites are, therefore, called geosynchronous satellites.
3. Hence, they are stationary with respect to Earth and can observe a specific portion of Earth continuously.
4. Geostationary satellites are not useful for studies of polar regions because they orbit only above the equator.

⭐ 3 Marks Questions: ⭐

1. Complete the table. [2025, 2021]

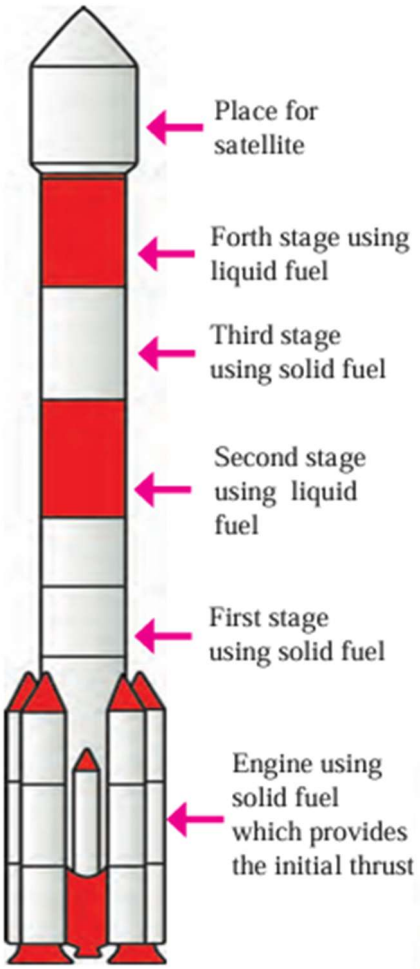
IRNSS	Fix the location of any place on Earth's surface in terms of precise latitude and longitude.	Navigational Satellite
INSAT and GSAT	Weather study & predict	Weather satellite
IRS	Study of forests, deserts, oceans, polar ice exploration and management of natural resources	Earth's observation

2. Why it is beneficial to use satellite launch vehicles made of more than one stage?

Ans:

1. The launch vehicle's structure and fuel are determined by the satellite's weight and orbit type.

2. Fuel forms a major part of the vehicle's total weight requiring the vehicle to carry a large fuel load.
3. To reduce weight multi-stage launch vehicles are used shedding stages as they ascend.
4. In 2 stage vehicle the first stage's fuel and engine propel it to a certain height and velocity.
5. Once the first stage fuel is exhausted its empty tank and engine are detached to reduce weight.
6. The second stage then ignites allowing the lighter vehicle to gain higher speeds.
   Therefore, it is beneficial to use satellite launch vehicles made of more than one stage.



10.5 a. Structure of PSLV made by ISRO

⭐ 5 Marks Questions: ⭐

1. What is meant by satellite launch vehicles? [2024]
Explain a satellite launch vehicle developed by ISRO with the help of a schematic diagram.

Ans:

Satellite launch vehicles are used, to place the satellites in their specific orbits. The functioning of the satellite launch vehicle is based on the Newton's third law of motion [2022] ... *+ refer previous answer.*

2. What is meant by an artificial satellite? How are the satellites classified based on their functions?

Ans:

A manmade object revolving around Earth or any other planet in a fixed orbit is called an artificial satellite. [2022, 2023, 2020]

Depending on their functions, satellites are classified into following categories:

1. **Weather satellite**: It is used to study and predict the weather. For example, INSAT and GSAT.
2. **Communication satellite**: It establishes communication between different locations in the world. For example, INSAT and GSAT.
3. **Broadcast satellite**: It is used for telecasting of television programs. For example, INSAT and GSAT.
4. **Navigational satellite**: It is used to locate the position of any place exactly on Earth's surface in terms of its precise latitude and longitude. For example, IRNSS.

5. **Military satellite**: It is used to collect information for security aspects. It provides information about the actions of the enemy through aerial surveillance.

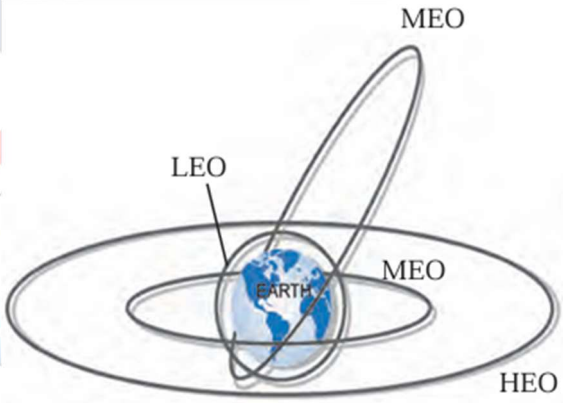
6. **Earth Observation Satellite**: It is used for the study of forests, deserts, oceans, polar ice on Earth's surface. It is also used for exploration and management of natural resources, observation and guidance, in case of natural calamities like floods and earthquakes.

3. What is meant by the orbit of a satellite? On what basis and how are the orbits of artificial satellites classified?

Ans:

The orbit of a satellite is the regular and the repeating path of the satellite, moving around an object in space. All artificial satellites revolve in different orbits.

Depending on the height of a satellite's orbit above Earth's surface, orbits of satellites are classified as below:



10.4 Orbits of satellites

1. **High Earth Orbit (HEO)**: [2025]
   - **i.** If the height of the satellite's orbit above Earth's surface is **greater than or equal to 35,780 km,** the orbit is called high Earth orbit.
   - **ii.** In these orbits, the satellite will take around **24 hours** to complete one revolution.
   - **iii.** Geosynchronous satellites revolve in HEOs and are used in applications like meteorology and for carrying signals for telephone, radio, etc.

2. **Medium Earth Orbit (MEO)**: [2023]
   - **i.** If the height of the satellite orbit above Earth's surface is **between 2,000 km and 35,780 km**, the orbit is called medium Earth orbit.
   - **ii.** The elliptical MEOs pass over the polar regions, so they are also called polar orbits. In these orbits, the satellites complete one revolution in **2 to 24 hours**.
   - **iii.** Global positioning satellites revolve in MEOs.

3. **Low Earth Orbit (LEO):**
   - **i.** If the height of the satellite orbit above Earth's surface is **between 180 km and 2,000 km**, the orbit is called low Earth orbit. [2022]
   - **ii.** Satellites used for scientific experiments and atmospheric studies revolve in LEOs.
   - **iii.** Depending on the height of their orbits, they complete one revolution in around **90 minutes**. International Space Station and Hubble telescope also revolve in LEOs.

II] Extra - Previous Years' Questions: (PYQs)

⭐ 1 Mark Objectives: ⭐

1. **Answer the following:**

1. Write the name of small satellite made by a group of students from COEP (College of Engineering, Pune) sent to the space through ISRO in 2016.
Ans: Swayam

2. Name the first satellite launched by Russia.
Ans: Sputnik

2. **Odd one out**: INSAT, GSAT, IRS, **PSLV** [2022]

⭐ 2 Marks Questions: ⭐

1. **The rocket is a type of firecracker used in Diwali.** [2025]

a. Name the launcher.
Ans: PSLV and GSLV are two important launchers

b. Name the law on which its working is based.
Ans: Its working is based on Newton's third law of motion.

2. **Give reason: It is necessary to manage space debris.** [2024]

Ans:

1. Space debris includes non-functional satellites; detached parts of launchers and pieces produced due to collisions in space.
2. According to an estimate made in 2016, about 2 crore pieces larger than 1 cm were revolving around the Earth.
3. Space debris can collide with artificial satellites and spacecraft, causing damage to them.
4. Space debris is increasing day by day, making it difficult to launch new spacecraft; therefore, it is essential to manage it.

⭐ 3 Marks Questions: ⭐

1. Complete the table: [2023, 2025]

Ans:

Type of Satellite	Function of the satellite	Satellite and launch vehicle
Navigational Satellite	Fix the location of any place on Earth's surface in terms of precise latitude and longitude	Satellites: **IRNSS** Launcher: **PSLV**
Earth Observation satellite	**Study of forests, deserts, oceans, polar ice exploration and management of natural resources**	Satellites: **IRS** Launcher: **PSLV**
Weather Satellite	**Weather study and predict**	Satellites: INSAT and GSAT Launcher: GSLV

2. Give the names of the following satellite series developed by India. [2024]

a. **Satellite series actively working in the field of telecommunication television and meteorological services.** **Ans**: INSAT and GSAT
b. **Satellite series especially working in the field of Education** **Ans**: EDUSAT
c. **Satellite series to exactly locate position of any place on Earth's surface.** **Ans:** IRNSS

3. What is the contribution of India in space technology? [2022]

Ans:

1. India also has made remarkable progress in the science and technology of launch vehicles.
2. India has developed various launch vehicles to place satellites weighing up to 2500 kg into different orbits. **PSLV and GSLV** are two important launch vehicles developed by India.
3. **INSAT and GSAT** satellites provide telecommunication, television broadcasting and meteorological services.
4. **EDUSAT** is specially used for educational purposes.
5. **IRS** satellites are used for monitoring and managing natural resources and disaster management.
6. **IRNSS** satellites are used to determine the precise latitude and longitude of any place on the Earth's surface.

4. State the importance of space missions. [2022]

Ans:

1. Space missions have made the world a global village through fast communication.

2. We can gather information about worldwide events and access information easily through the internet.
3. Satellites help in providing advance warnings about natural calamities and taking proper precautions.
4. Satellites help in aerial surveillance to obtain information about enemy activities during wars.
5. Space technology helps in exploring fossil reserves and minerals present in the Earth.
6. Space technology has become an essential part of the development of a nation.

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